

Planetary Science & Solar System Dynamics

just a bunch of orbital yeets

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Prerequisite Knowledge

- Multi-wavelength blackbody flux conversions
- The vis-viva equation and tidal acceleration scaling
- Single-variable optimization (setting a derivative to zero and solving)
- Two-dimensional vector kinematics (adding velocities as vectors, not just speeds)
- Braincells (still required, now with orbital velocity)

1 Introduction

Every planet, moon, comet, and speck of dust in the solar system is playing the same single game: falling toward the Sun, and continuously missing. Everything in this handout is really just bookkeeping (yes, I like this word if you haven't figured it out already) on that one theme — how much energy a body has, how it's shared between orbital motion and heat, and how spacecraft can borrow a little bit of that bookkeeping from the planets themselves to get around for free. We'll start at the center of it all, work our way out through the birth of the planets, then treat the planets as thermodynamic objects, and finally treat them as obstacles and tools for getting a spacecraft from one place to another.

2 The Central Powerhouse: The Sun as an Engine

2.1 Solar Internal Architecture

The Sun is not a uniform ball of gas — it's stratified into layers, each defined by *how energy moves through it*, not by composition.

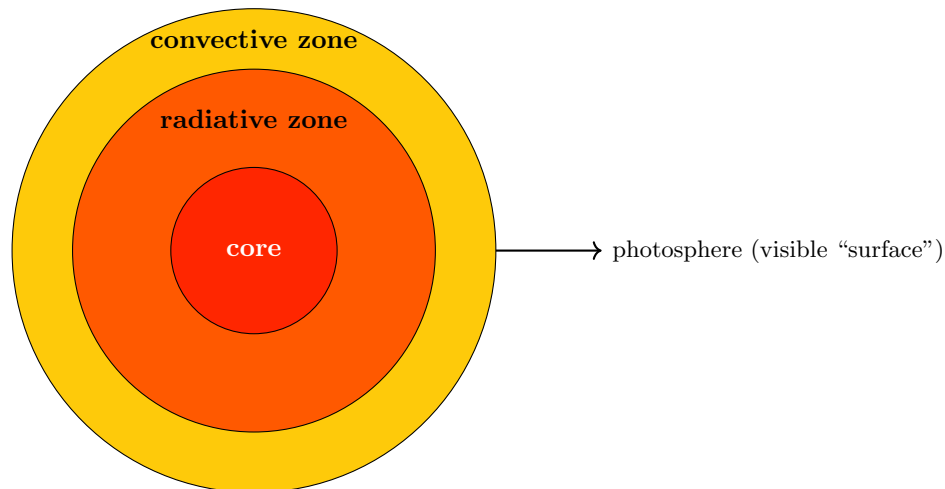


Figure 1: Cross-section of the Sun's interior, layered by energy-transport mechanism rather than by material.

- **Core** ($r \lesssim 0.25R_{\odot}$): temperature $\sim 1.5 \times 10^7 K$, density $\sim 150 g/cm^3$. Hot and dense enough to sustain the proton-proton chain, fusing hydrogen into helium and releasing the energy that powers everything above it.
- **Radiative zone** ($0.25\text{--}0.7R_{\odot}$): so dense that photons cannot simply stream outward. Instead, each photon is absorbed and re-emitted in a random direction over and over, performing a **random walk** outward. This is why energy generated in the core today takes on the order of $10^4\text{--}10^5$ years to reach the surface, even though light itself only needs about 2 seconds to cross the Sun's radius.
- **Convective zone** ($0.7R_{\odot}$ to the surface): the temperature gradient becomes steep enough that bulk fluid motion — hot plasma physically rising, cooling, and sinking again — transports energy faster than radiative diffusion can. This is the same instability that makes a pot of water roll when heated from below, just on a stellar scale, and it's what produces the mottled “granulation” visible on the solar surface.

- **Photosphere:** the thin layer ($\sim 500\text{ km}$) where the plasma finally becomes transparent enough that photons escape freely into space. This is the layer we mean when we say “the surface of the Sun” and the layer whose temperature ($\sim 5778\text{ K}$) we use in blackbody calculations.

2.2 The Solar Constant

The Sun radiates as a blackbody, with total luminosity fixed by the Stefan-Boltzmann law applied at the photosphere:

$$L_{\odot} = 4\pi R_{\odot}^2 \sigma T_{eff}^4.$$

That energy then spreads out over ever-larger spheres as it travels outward, so the flux measured at any distance d is just that same luminosity divided by the surface area of a sphere of radius d :

$$S(d) = \frac{L_{\odot}}{4\pi d^2}.$$

Evaluating this at $d = 1\text{ AU}$ defines the **solar constant**:

$$S_{\odot} = \frac{(4\pi R_{\odot}^2 \sigma T_{eff}^4)}{4\pi (1\text{ AU})^2} = \sigma T_{eff}^4 \left(\frac{R_{\odot}}{1\text{ AU}} \right)^2 \approx 1361\text{ W/m}^2.$$

Notice the elegant simplification: the 4π terms cancel entirely, leaving the solar constant as just the photosphere’s own emitted flux (σT_{eff}^4), diluted purely by the geometric factor $(R_{\odot}/d)^2$ — the same inverse-square dilution that shows up everywhere in this handout.

2.3 Space Weather: Solar Wind, Heliosphere, and Magnetosphere

The Sun doesn’t just emit light — it constantly sheds charged particles (mostly protons and electrons) outward as the **solar wind**, moving fast enough ($\sim 400\text{ km/s}$ typically) to escape the Sun’s gravity entirely. This wind inflates a vast bubble around the entire solar system called the **heliosphere**; its outer edge, the **heliopause**, is where the solar wind’s outward pressure finally drops below the pressure of the interstellar medium pushing back.

A planet with its own internal magnetic field — Earth, most obviously — carves out a smaller bubble of its own within the solar wind, called the **magnetosphere**. The boundary where the planet’s magnetic pressure balances the incoming solar wind’s ram pressure is the **magnetopause**. This balance is precisely what shields a planet’s atmosphere from being slowly stripped away by the solar wind — widely believed to be a major reason Mars, which lost its global magnetic field billions of years ago, also lost most of its atmosphere, while Earth kept both.

2.4 The Solar Neutrino Anomaly

The proton-proton chain in the core produces not just photons but also **neutrinos** — nearly massless particles that interact so weakly with matter that they stream straight out of the Sun in about 2 seconds, completely bypassing the photon’s 10^4 -year random walk. That makes solar neutrinos a uniquely direct, real-time probe of what’s happening in the core *right now*, unlike sunlight, which tells us about fusion that happened millennia ago.

When detectors first counted solar neutrinos in the 1960s, they found only about a third of the number predicted by fusion models — the **solar neutrino problem**. The resolution turned out

to be amazingly profound: neutrinos come in three “flavors,” and as they travel they continuously oscillate between them (**neutrino oscillation**), a phenomenon that also requires neutrinos to have a small but nonzero mass. Early detectors were only sensitive to one flavor, so they were, quite literally, only counting a third of the particles that were actually there. This was less a crisis for stellar physics than a landmark discovery in particle physics.

2.5 Radiation Pressure

Light carries momentum, not just energy: a photon of energy E carries momentum $p = E/c$. A surface that absorbs a flux S (energy per area per time) is therefore absorbing momentum at a rate S/c per unit area per unit time — which is exactly a pressure:

$$P_{rad} = \frac{S}{c}.$$

For an object of cross-sectional (absorbing) area σ sitting a distance d from a star of luminosity L , the total *force* from radiation pressure is

$$F_{rad} = \frac{L\sigma}{4\pi d^2 c}.$$

Problem: The Radiation-Gravity Balance Radius

A spherical dust grain of radius r and density ρ orbits a star of mass M and luminosity L at distance d . Treating the grain as a perfect absorber with geometric cross section $\sigma = \pi r^2$, find the critical radius r_c below which radiation pressure exceeds gravity.

Gravity pulls inward with $F_{grav} = GMm/d^2$, where $m = \frac{4}{3}\pi r^3 \rho$ is the grain’s mass. Radiation pressure pushes outward with $F_{rad} = L\pi r^2/(4\pi d^2 c)$. Setting them equal:

$$\frac{GM \left(\frac{4}{3}\pi r^3 \rho\right)}{d^2} = \frac{L\pi r^2}{4\pi d^2 c}.$$

Both sides scale as $1/d^2$ — the distance cancels entirely, which makes sense: both forces dilute with the same inverse-square law, so whichever one wins, wins everywhere. Solving for r :

$$r_c = \frac{3L}{16\pi GM\rho c}.$$

Grains smaller than r_c get blown out of the system by radiation pressure faster than gravity can hold them (this is one mechanism, alongside the velocity-dependent Poynting-Robertson drag, that keeps interplanetary dust disks from simply lasting forever).

3 Solar System Formation and Boundary Zones

3.1 The Nebular Hypothesis and Angular Momentum

The Sun and planets formed from a single collapsing cloud of gas and dust: the **solar nebula**. As the cloud contracted under its own gravity, any small initial rotation was *amplified*, exactly the

way a spinning ice skater speeds up when pulling their arms in — angular momentum $L = I\omega$ is conserved, so shrinking the moment of inertia I forces ω up.

That spin-up has an unavoidable geometric consequence: material can collapse freely along the rotation axis (nothing opposes it), but material moving perpendicular to the axis is held up by its own angular momentum, resisting collapse in that direction. The result is that a roughly spherical cloud flattens into a thin, rapidly-rotating **protoplanetary disk**, with the young Sun forming at its center. Every planet in the solar system orbiting in almost the same plane, and in the same direction, is a direct fossil record of this one flattening event.

3.2 The Snow Line

Close to the young Sun, temperatures in the disk were too high for volatile compounds like water, ammonia, and methane to condense into solid ice — they stayed as gas. Farther out, past a critical distance called the **snow line** (or frost line), the disk was cold enough for these volatiles to freeze into solid ice grains, dramatically increasing the amount of solid material available for building planets.

This single boundary explains the solar system’s most basic architecture:

- **Inside the snow line:** only rocky and metallic material could condense, so planets that formed here (Mercury through Mars) ended up small and dense — the **terrestrial planets**.
- **Beyond the snow line:** ice added enormously to the available solid mass, letting protoplanetary cores grow large enough, fast enough, to gravitationally capture huge envelopes of hydrogen and helium gas directly from the surrounding nebula before it dissipated — producing the **Jovian (gas/ice giant) planets**.

3.3 The Outer Reaches: Kuiper Belt, Oort Cloud, and Dwarf Planets

- **Kuiper Belt:** a flattened, disk-shaped belt of icy debris beyond Neptune ($\sim 30\text{--}50\text{ AU}$), left over from planet formation — the source of most short-period comets and home to Pluto, Eris, and other trans-Neptunian objects.
- **Oort Cloud:** a roughly spherical shell of icy bodies far beyond the Kuiper Belt ($\sim 10^3\text{--}10^5\text{ AU}$), thought to be the reservoir for long-period comets, which arrive from essentially random directions on the sky — consistent with a spherical, not disk-shaped, source.
- **Dwarf planets:** the IAU’s 2006 definition requires a body to (i) orbit the Sun, (ii) be massive enough for self-gravity to pull it into a round shape (hydrostatic equilibrium), but (iii) have *not* cleared its orbital neighborhood of other debris. That third criterion is exactly what demotes Pluto: it shares its orbital zone with the rest of the Kuiper Belt, whereas the eight “full” planets have gravitationally swept their orbits nearly clean.

4 Planetary Energy Budgets: The Thermodynamics of Worlds

4.1 Albedo: Bond vs. Geometric

Not all incoming sunlight gets absorbed — some fraction is reflected straight back, and that fraction is called the **albedo**. There are two distinct definitions in use, and olympiad problems love to test the difference:

- **Bond albedo** (A): the fraction of *total* incident sunlight, integrated over *all* wavelengths and *all* reflection angles, that gets reflected back to space. This is the physically fundamental quantity — it’s what actually enters the energy balance in Section 4.2.
- **Geometric albedo** (p): the brightness of a planet at zero phase angle (fully illuminated, as seen from the Sun), compared to the brightness of a flat, perfectly diffuse (Lambertian) disk of the same size at the same distance. This is an *observational* quantity, measured by astronomers pointing a telescope at a planet, and it says nothing directly about total absorbed energy, since it only samples one reflection geometry.

A planet can have a geometric albedo greater than 1 (some icy moons do) if it happens to reflect unusually efficiently straight back toward the direction of illumination, even while its Bond albedo — the actual fraction of total energy reflected in all directions — stays comfortably below 1.

4.2 Effective Temperature

Think of a planet as a simple energy-in, energy-out system. **Energy in:** the star illuminates only the planet’s cross-sectional disk, πR_p^2 , and a fraction $(1 - A)$ of that flux is actually absorbed (the rest reflects away per the Bond albedo). **Energy out:** the planet re-radiates as a blackbody from its *entire* surface area, $4\pi R_p^2$.

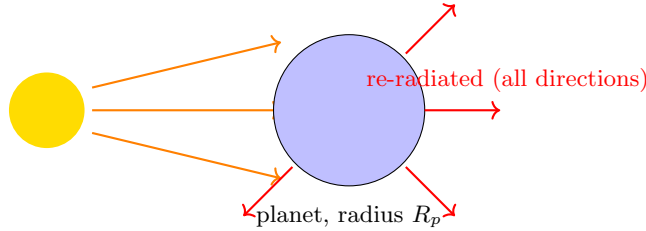


Figure 2: Energy balance for a planet. Absorption only “sees” the disk πR_p^2 facing the star; emission comes from the full surface $4\pi R_p^2$.

Setting absorbed power equal to emitted power at equilibrium:

$$\underbrace{\frac{L(1-A)}{4\pi d^2} \pi R_p^2}_{\text{absorbed}} = \underbrace{4\pi R_p^2 \sigma T_{eff}^4}_{\text{emitted}}.$$

The R_p^2 and one factor of π cancel from both sides — the planet’s own size drops out entirely, which makes sense, since a bigger planet both catches and radiates proportionally more energy. Solving for temperature:

$$T_{eff} = \left[\frac{L(1-A)}{16\pi d^2 \sigma} \right]^{1/4}.$$

Note this is the temperature the planet *would* have with no atmosphere at all; any real greenhouse effect (Section 4.3) sits on top of this baseline.

4.3 The Greenhouse Scaffolding

A planet’s atmosphere can be modeled, to a useful first approximation, as a stack of thin layers, each of which is perfectly transparent to visible sunlight (letting it pass straight through to be absorbed at the surface) but perfectly opaque to the infrared radiation the warm surface emits

back upward. Each such layer absorbs all the infrared it receives and, being in its own equilibrium, must re-radiate that same power — but it does so from *both* its faces, sending half back down and half further up.

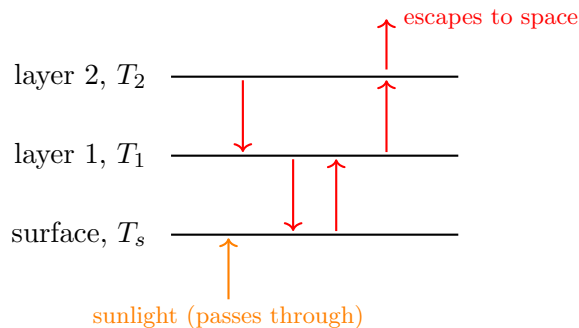


Figure 3: A two-layer greenhouse model. Sunlight passes straight through every layer; each layer re-radiates absorbed infrared equally up and down.

Work outward from the top layer down. In steady state, whatever power ultimately escapes to space from the outermost layer must equal the planet’s effective (no-atmosphere) emission, σT_{eff}^4 , since energy in must still equal energy out overall. But every layer beneath radiates *that same power* out its top face too (each layer is itself in equilibrium with what it receives from below), while also sending an equal amount back down. Tracking the power balance layer by layer, one finds that each additional layer beneath the top adds exactly one more unit of σT_{eff}^4 to what the surface itself must radiate to stay in balance. For n layers total,

$$T_s^4 = (n + 1) T_{eff}^4.$$

A single layer ($n = 1$) already doubles the fourth power of temperature the surface must reach — this simple scaffolding, crude as it is, is enough to get the qualitative direction (and rough magnitude) of greenhouse warming right.

4.4 Tidal Dissipation and Orbital Resonances

A moon on an eccentric orbit is squeezed and stretched by a different amount at every point in its orbit (Handout 2 covered why tidal acceleration scales as $1/d^3$, so the tidal bulge is stronger at perigee than at apogee). If the moon’s rotation is tidally locked (synchronous), this flexing amplitude cannot simply vanish — the shape the moon “wants” to relax into changes continuously as it moves around its eccentric orbit, forcing its interior to physically flex on every orbit. That flexing is not perfectly elastic: friction inside the moon converts some of that mechanical energy directly into heat. A standard (if simplified) expression for this dissipated power is

$$\dot{E}_{tidal} = \frac{21}{2} \frac{k_2}{Q} \frac{GM_p^2 R^5 n e^2}{a^6},$$

where k_2 is the moon’s Love number (how readily it deforms), Q is its tidal dissipation quality factor (how lossy that deformation is), R and a are the moon’s radius and orbital semi-major axis, M_p is the parent planet’s mass, and $n = 2\pi/P$ is the mean motion. The crucial takeaway is the e^2 dependence: with zero eccentricity, there is no time-varying flexing at all, and the heating vanishes.

This raises an obvious puzzle for Io, Jupiter’s innermost large moon: tidal forces from Jupiter should, over time, damp Io’s eccentricity to zero (tides always act to circularize orbits), which should shut off the heating entirely — yet Io is observed to be the most volcanically active body in the solar system. The resolution is the **Laplace resonance**: Io, Europa, and Ganymede orbit in a 4:2:1 mean-motion resonance, meaning their mutual gravitational tugs repeat at the same orbital phase every cycle, continuously pumping Io’s eccentricity back up even as Jupiter’s tides try to damp it out. The resonance and the tidal dissipation are locked in a permanent tug of war, and Io’s volcanism is the heat generated by that ongoing fight.

5 Orbital Interplanetary Maneuvers

5.1 The Hohmann Transfer Ellipse

The cheapest (lowest fuel) way to move a spacecraft between two circular, coplanar orbits is the **Hohmann transfer**: an ellipse tangent to the inner circular orbit at its perigee and to the outer circular orbit at its apogee.

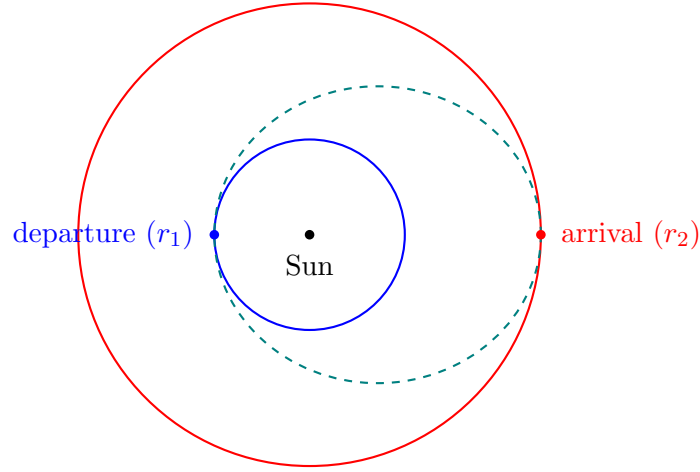


Figure 4: The Hohmann transfer ellipse (teal, dashed), tangent to the inner circular orbit (blue) at perigee and the outer circular orbit (red) at apogee.

The transfer ellipse has semi-major axis $a_t = (r_1 + r_2)/2$. Using the vis-viva equation, $v^2 = GM \left(\frac{2}{r} - \frac{1}{a} \right)$, evaluated on the circular orbits ($a = r$) and on the transfer ellipse ($a = a_t$) at each radius:

$$\begin{aligned} v_{c,1} &= \sqrt{\frac{GM}{r_1}}, & v_p &= \sqrt{GM \left(\frac{2}{r_1} - \frac{1}{a_t} \right)}, & \Delta v_1 &= v_p - v_{c,1}, \\ v_{c,2} &= \sqrt{\frac{GM}{r_2}}, & v_a &= \sqrt{GM \left(\frac{2}{r_2} - \frac{1}{a_t} \right)}, & \Delta v_2 &= v_{c,2} - v_a. \end{aligned}$$

Both burns are prograde speed changes (tangent to the orbit, in the direction of motion) — the first speeds the spacecraft up onto the transfer ellipse, and the second speeds it up again (for an outbound transfer, where $v_a < v_{c,2}$) to circularize once it arrives. The total mission cost is simply

$$\boxed{\Delta v_{total} = |\Delta v_1| + |\Delta v_2|}$$

Since the transfer orbit is just half of a full ellipse, the trip takes exactly half the transfer orbit's own Keplerian period:

$$t_{transfer} = \pi \sqrt{\frac{a_t^3}{GM}}.$$

5.2 Synodic Periods

A Hohmann transfer only works if the destination planet actually shows up at the rendezvous point at the right time — you can't just launch whenever you like. The relevant timescale is the **synodic period**: how often two planets return to the same relative configuration as seen from the Sun (e.g., Earth back at opposition with Mars).

Each planet sweeps out angle at its own constant rate $2\pi/P$. The *relative* angular speed between two planets is the difference of their individual rates, so the time for that relative angle to complete a full 2π lap (returning to the same configuration) is

$$\frac{1}{S} = \left| \frac{1}{P_1} - \frac{1}{P_2} \right|.$$

Note that for two planets with nearly equal periods, S blows up — their relative configuration barely changes from year to year, so it takes an enormous time for them to realign.

5.3 Interception Geometry: Phase Angles

Launch windows open only once per synodic period, but even within a synodic cycle, you need the target planet positioned correctly *at launch* so that it arrives at the transfer orbit's apogee at exactly the moment your spacecraft does.

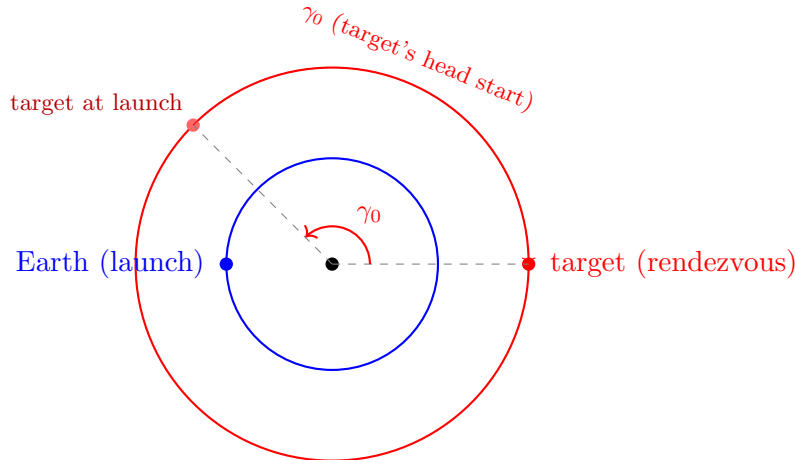


Figure 5: The target planet must have a head-start phase angle γ_0 at launch, so that its own steady motion carries it to the rendezvous point exactly as the spacecraft arrives.

During the transfer time $t_{transfer}$ (Section 5.1), the target planet sweeps through an angle $\omega_2 t_{transfer} = \frac{360^\circ}{P_2} t_{transfer}$. Since the spacecraft itself always travels exactly 180° around the Sun on a Hohmann transfer (it's half of an ellipse), the target must *already* be short of the 180° rendezvous point, by

exactly the angle it will cover during the flight:

$$\gamma_0 = 180^\circ - \frac{360^\circ}{P_2} t_{transfer}.$$

If γ_0 comes out negative, that simply means the target needs to *lag behind* Earth (for an outer planet, which moves slower and so falls behind less than 180° over the flight time – this sign flips depending on whether the destination is interior or exterior, so always sanity check the geometry rather than trusting the sign blindly).

6 Gravity Assists and Hyperbolic Flybys

6.1 Hyperbolic Trajectories

If a spacecraft passes a planet with enough relative speed that its energy relative to that planet is positive, its trajectory (in the planet's own reference frame) is an unbound hyperbola rather than an ellipse. Far from the planet, the spacecraft approaches and recedes along straight asymptotes at the same constant speed, called the **hyperbolic excess velocity** v_∞ (the speed it would have left with even if the planet weren't there to redirect it). The one thing the flyby does actually change, in the planet's frame, is the *direction* of that velocity vector — by a turning angle δ that depends on how close the approach is (the impact parameter b):

$$\tan\left(\frac{\delta}{2}\right) = \frac{GM_p}{b v_\infty^2}.$$

A very distant flyby (large b) barely deflects the spacecraft at all ($\delta \rightarrow 0$); a very close flyby can, in principle, swing the trajectory around by nearly 180° .

6.2 Heliocentric Momentum Stealing

Here's the trick that makes gravity assists useful: the flyby only rotates the spacecraft's velocity *in the planet's own moving reference frame* — it does not change the *speed* in that frame at all ($|\vec{v}_{in}| = |\vec{v}_{out}| = v_\infty$, energy conservation in the planet's frame). But the planet's frame is itself moving relative to the Sun at velocity $\vec{v}_p$, and when you translate back into the Sun's frame, that rotation of direction can add constructively or destructively with $\vec{v}_p$.

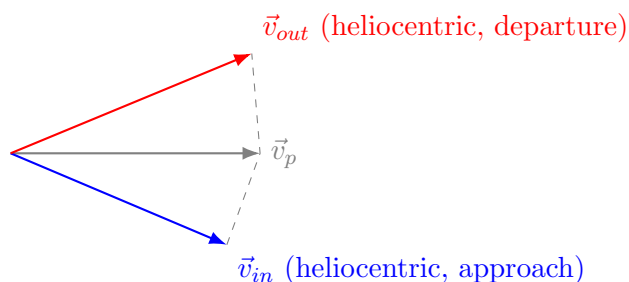


Figure 6: Vector construction of a gravity assist. The planet's own velocity $\vec{v}_p$ (gray) doesn't change; the spacecraft's velocity relative to the planet gets rotated by δ , and adding $\vec{v}_p$ back on can leave the spacecraft with more (or less) heliocentric speed than it arrived with.

For the ideal case of a trailing-side flyby (where the rotation swings the relative-velocity vector to align as closely as possible with $\vec{v}_p$), the heliocentric speed boost approaches

$$\Delta v_{helio} \approx 2v_{\infty} \sin\left(\frac{\delta}{2}\right).$$

Crucially, this energy isn't free in any absolute sense — it's stolen directly from the planet's own orbital motion. The planet slows down (or speeds up) by an utterly imperceptible amount in exchange, because its mass so vastly exceeds the spacecraft's that momentum conservation splits the exchange almost entirely in the spacecraft's favor.

6.3 Tisserand's Parameter

In the restricted three-body problem (Sun, a massive planet, and a test-particle spacecraft or comet), a single combination of orbital elements — semi-major axis a , eccentricity e , and inclination i relative to the perturbing planet's own orbit at a_J — stays very nearly constant across an encounter, even though a , e , and i individually can each change dramatically:

$$T_J = \frac{a_J}{a} + 2\sqrt{\frac{a}{a_J}(1-e^2)\cos i}.$$

This conserved quantity, **Tisserand's parameter**, emerges from the Jacobi constant of the restricted three-body problem (itself a conserved quantity in the rotating frame where the planet is stationary), re-expressed in terms of ordinary heliocentric orbital elements. Its practical use is as a fingerprint: if a newly observed comet has almost the same T_J as a previously known object, that's strong evidence they're the *same* body, just with its orbit reshuffled by an intervening close encounter with Jupiter (this is exactly how several “Jupiter-family comets” have been identified as returning objects rather than new discoveries).

7 Planetary Phenomena and Sight Geometry

7.1 Apparent Retrograde Motion

Every planet moves steadily eastward against the background stars (**prograde** motion) on average — yet for weeks at a time, outer planets appear to reverse and drift westward (**retrograde** motion) before resuming their normal course. This is a pure illusion of relative motion, not a real reversal of any planet's orbit.

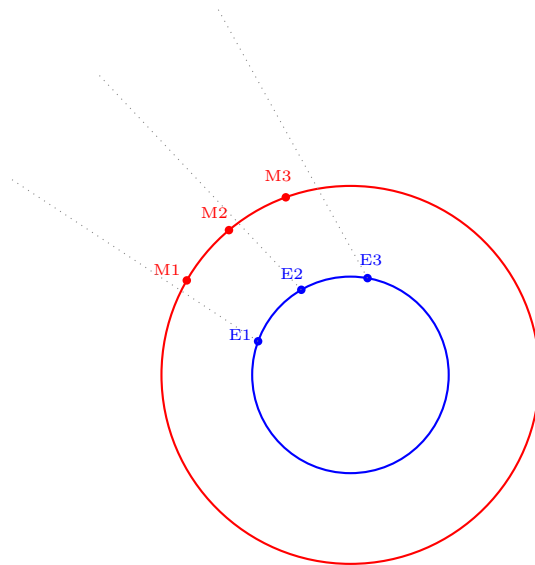


Figure 7: Earth (inner, blue) moves faster than an outer planet (red) and overtakes it near opposition. The sight-line from Earth to the planet (dotted) briefly sweeps backward against the stars during the overtaking, even though both planets keep moving prograde around the Sun.

By Kepler’s third law, the inner planet (Earth) always moves faster in its orbit than any outer planet. Near **opposition** (when Earth is directly between the Sun and the outer planet), Earth is momentarily overtaking the slower outer planet “on the inside track,” the way a faster car passing a slower one on an adjacent lane briefly appears, from the slow car’s own point of view, to move backward. The line of sight from Earth to the planet sweeps westward relative to the stars during this passing maneuver, even though neither planet’s own orbital motion ever reverses.

7.2 Elongations, Quadratures, Oppositions, and Phase Functions

- **Elongation:** the angular separation, as seen from Earth, between a planet and the Sun. For an inferior planet (Mercury, Venus), this angle is bounded — it can never exceed a **greatest elongation**, since the planet’s own orbit is entirely inside Earth’s.
- **Conjunction:** Sun and planet share the same geocentric direction (elongation 0°); **inferior conjunction** for an inferior planet passing between Earth and Sun, **superior conjunction** for one passing behind the Sun.
- **Quadrature:** applies only to superior planets (Mars and beyond) — elongation exactly 90° , the planet appears at a right angle from the Sun in the sky.
- **Opposition:** applies only to superior planets — elongation 180° , the planet is opposite the Sun in the sky, rising as the Sun sets. This is also, not coincidentally, the closest and brightest a superior planet ever gets, since it’s the point of nearest approach to Earth.

The fraction of a planet’s visible disk that is illuminated depends on the **phase angle** α (Sun-planet-Earth angle) via

$$k = \frac{1 + \cos \alpha}{2}.$$

An inferior planet, viewed from Earth, therefore cycles through phases just like the Moon (new, crescent, quarter, gibbous, full) as its phase angle sweeps through a full range; a superior planet’s

phase angle is always fairly small (since Earth’s orbit is the smaller of the two), so it never appears less than about “gibbous” from Earth.

7.3 Libration

The Moon’s rotation period exactly matches its orbital period (tidal locking), which is often described as meaning we only ever see “one side” of the Moon. In reality we see slightly more than half — about 59% of the total surface over time — because of **libration**.

Libration in longitude arises because the Moon’s rotation proceeds at a constant angular rate, while its orbital angular speed (by Kepler’s second law) speeds up near perigee and slows down near apogee. The two rates only match *on average*; at any other moment they’re slightly out of step, so we get to peek a little around the eastern or western limb depending on where the Moon is in its elliptical orbit. To leading order in eccentricity, the amplitude of this optical libration in longitude is

$$\ell_{max} \approx 2e \text{ (radians),}$$

directly inherited from the same equation-of-center correction that describes any body’s angular position on an eccentric orbit. There is a second, smaller effect, **libration in latitude**, caused by the small ($\sim 6.7^\circ$) tilt of the Moon’s rotation axis relative to its orbital plane, letting us alternately peek over its north and south poles.

7.4 Shadow Cones: Umbra and Penumbra

When a planet or moon blocks sunlight, it casts two distinct shadow regions: the **umbra**, where the Sun’s disk is completely blocked, and the **penumbra**, where it’s only partially blocked.

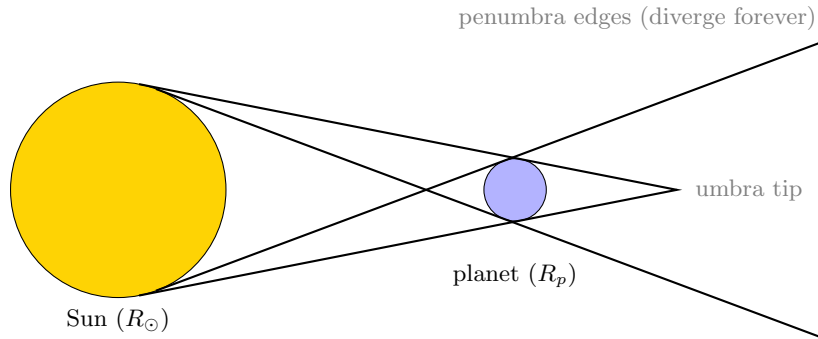


Figure 8: The umbra (solid lines) is the fully-dark cone that converges to a point, since the planet is smaller than the Sun. The penumbra (the region between the solid and extended dashed lines) diverges outward without ever closing.

The umbra is a converging cone precisely because the Sun’s disk is larger than the planet’s — similar triangles relate its length L_u (measured from the planet) to the planet-Sun distance d :

$$L_u = \frac{d R_p}{R_\odot - R_p}.$$

If an observer’s distance from the planet exceeds L_u , they can no longer be fully shadowed by the umbra at all — this is exactly the geometry behind an **annular** solar eclipse, where the Moon’s umbra falls just short of reaching Earth’s surface, leaving a thin ring of the Sun visible around the Moon’s silhouette.

8 Summary Sheet

Topic	Result
Solar constant	$S(d) = L_{\odot}/(4\pi d^2)$ at distance d
Radiation pressure	$F_{\text{rad}} = L\sigma/(4\pi d^2 c)$ for cross section σ
Rad.-gravity balance	$r_c = 3L/(16\pi GM\rho c)$ (dust grain size limit)
Planet temperature	$T_{\text{eff}} = [L(1 - A)/(16\pi d^2 \sigma_{\text{SB}})]^{1/4}$
Greenhouse model	$T_s^4 = (n + 1)T_{\text{eff}}^4$ (n -layer atmosphere)
Tidal heating	$\dot{E}_{\text{tidal}} = \frac{21}{2} \frac{k_2}{Q} \frac{GM_p^2 R^5 n e^2}{a^6}$
Hohmann Δv	$v^2 = GM(2/r - 1/a_t)$ where $a_t = (r_1 + r_2)/2$
Hohmann duration	$t_{\text{transfer}} = \pi \sqrt{a_t^3/GM}$ (half period)
Synodic period	$1/S = 1/P_1 - 1/P_2 $
Launch phase angle	$\gamma_0 = 180^\circ - (360^\circ/P_2) t_{\text{transfer}}$
Flyby turning angle	$\tan(\delta/2) = GM_p/(bv_{\infty}^2)$
Gravity assist	$\Delta v_{\text{helio}} \approx 2v_{\infty} \sin(\delta/2)$ (max boost)
Tisserand's parameter	$T_J = a_J/a + 2\sqrt{(a/a_J)(1 - e^2)} \cos i$
Illuminated fraction	$k = (1 + \cos \alpha)/2$ for phase angle α
Optical libration	$\ell_{\text{max}} \approx 2e$ rad (amplitude in longitude)
Umbra length	$L_u = dR_p/(R_{\odot} - R_p)$